



Enhancing Multimodal Recommendation via Multimodal Representation Calibration in Spectral Domain

Minghui Wang

School of Artificial Intelligence, Engineering Research
Center of Knowledge-Driven Human-Machine Intelligence
of Ministry of Education, Jilin University
Changchun, China
wangmh24@mails.jlu.edu.cn

Yu Li*

School of Artificial Intelligence, Engineering Research
Center of Knowledge-Driven Human-Machine Intelligence
of Ministry of Education, Jilin University
Changchun, China
liyu90@jlu.edu.cn

Tingting Zhang

Institute of Big Data and Interdisciplinary Science,
School of Statistics and Data Science,
Jilin University of Finance and Economics
Changchun, China
103069@jlufe.edu.cn

Yi Chang

School of Artificial Intelligence, Engineering Research
Center of Knowledge-Driven Human-Machine Intelligence
of Ministry of Education, Jilin University
Changchun, China
yichang@jlu.edu.cn

Abstract

Multimodal recommendation, which integrates user-item interactions with rich item multimodal features to infer user preferences, has emerged as one of the state-of-the-art recommendation paradigms. However, most existing multimodal recommendation methods focus solely on the multimodal fusion process of items, overlooking the impact of modality-specific noise characteristics and inter-modal distribution shifts on the fusion process. As a result, noise from specific modalities may be inadvertently amplified during the fusion of multimodal representations, thereby degrading the quality of fused representations and ultimately undermining recommendation accuracy. To address this issue, we propose a novel framework, DAMPS, which **D**enoises and **cA**librates multimodal representations in the spectral domain from the perspectives of both **aM**plitude and **P**hase **S**pectrum. Specifically, we first apply the Fast Fourier Transform to decompose multimodal representations into amplitude spectrum and phase spectrum components. We then propose the adaptive phase calibrator, which resolves the phase distribution shift between modalities through the optimal minimum variance unbiased estimate of phase offset and adaptive phase rotation. We further develop the amplitude variance-ratio filter, which suppresses modality-specific amplitude noise while preserving prominent modality-specific representations. Finally, we introduce the inter-modal coherence filter, which quantifies and enhances the intrinsic shared information between modalities through amplitude-squared coherence, thereby eliminating irrelevant modality noise. Technically, DAMPS is model-agnostic and highly flexible, enabling seamless integration with various multimodal recommendation backbones. Extensive evaluations on five public multimodal recommendation datasets demonstrate that

DAMPS consistently improves recommendation performance and achieves state-of-the-art results.

CCS Concepts

• Information systems → Recommender systems.

Keywords

Recommender Systems, Multimodal Recommendation, Multimodal Representation Calibration, Model Agnostic

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1 Introduction

Modern recommendation systems primarily revolve around users' historical interactions, deeply mining latent patterns by modeling behaviors such as clicks and purchases to learn implicit user preferences and generate personalized recommendations [14, 16, 17, 30]. However, relying solely on the interaction data often fails to fully capture item characteristics or user intent. Consequently, multimodal information such as images, text, and even audio has been introduced to enrich item profiles. Recent studies have shown that such multimodal item information possesses inherent complementarity in the semantic space, capable of revealing attributes like color, style, and functionality at a fine-grained level, thus providing more precise contextual cues into user profiles [1, 46]. With the continuous advancement of multimodal fusion techniques, multimodal recommendation has demonstrated significant advantages in enhancing user and item representations, mitigating cold-start issues,

*Corresponding author.



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and improving interpretability, gradually emerging as a mainstream paradigm in recommendation systems [4, 19, 22, 23].

Multimodal recommendation systems prioritize projecting multimodal information such as visual, textual, and behavioral representations into a unified semantic space to ensure coherence in inter-modal interactions [13]. Early studies typically fuse multimodal representations through simple concatenation or addition operations [13, 24], effectively alleviating long-standing challenges such as the cold start issues that have plagued traditional recommendation methods [21, 36], and demonstrating the benefits of multimodal fusion [34, 37, 44, 48]. Building on these foundations, more sophisticated methods introduce attention mechanisms and dynamic gating to selectively emphasize complementary information from each modality, thereby enhancing representation consistency and robustness [41, 42, 49]. The latest line of work [26] extends this paradigm by exploring frequency-domain multimodal fusion. This approach employs Fourier spectrum filters to adaptively denoise and fuse multimodal representations, offering a new pathway to disentangle informative patterns from noise.

From the perspective of signal processing, multimodal representation fusion essentially corresponds to the process of multi-signal fusion. For a signal, in the spectral domain, the amplitude spectrum reflects the energy magnitude of each frequency component, while the phase spectrum encodes the relative spatial shifts of those components. However, existing multimodal recommendation methods either do not consider modality representation denoising at all, or rely solely on coarse, amplitude-based denoising techniques, overlooking modality-specific noise characteristics and phase distribution shifts between modalities in multimodal fusion. In real-world scenarios, multimodal representations are inherently heterogeneous. For instance, item images often contain irrelevant or distracting backgrounds, while text frequently includes redundant or ambiguous phrases. When these subtle yet pervasive noises are left untreated, multimodal representation fusion may inadvertently amplify modality-specific noise patterns, thereby distorting inter-modal interactions and contaminating the fused representations. Furthermore, overlooking the distribution shifts of the phase spectrum between modalities may lead to implicit semantic discrepancies, weaken semantic consistency, and ultimately undermine the recommendation performance.

To intuitively demonstrate the differences between image and text modalities, we visualize the spectral domain distributions of the initial image and text representations generated by MGCN [41] on the Amazon Clothing dataset, as shown in Figure 1. To quantitatively characterize these observations, we employ three complementary signal processing metrics: (1) *Spectral Entropy*, measuring the randomness of spectral distributions (lower entropy values indicate reduced noise and higher stability); (2) *Energy Ratio*, indicating the proportion of spectral energy residing within informative frequency bins (higher ratios represent richer semantic content); and (3) *Variance of Phase Offset*, measuring inter-modal consistency in the phase spectrum (smaller values indicate stronger phase synchronization between modalities). For detailed information about these metrics, please refer to Appendix A. It can be observed that the raw multimodal representations exhibit higher Spectral Entropy values ($\mu_{\text{image}} = 5.03$, $\mu_{\text{text}} = 4.37$), indicating substantial spectral randomness, with approximately 32.4% of modality features dominated by

noise. Furthermore, their Energy Ratios are relatively low (0.56 for images, 0.77 for text), implying that roughly 33.5% of the spectral energy corresponds to uninformative or irrelevant features. Critically, the abnormally large inter-modal phase variance indicates significant statistical asynchrony, and such inconsistency may lead to information loss during multimodal representation fusion. These findings demonstrate that without targeted spectral denoising and phase calibration strategies, multimodal recommendation frameworks inevitably generate unstable and semantically fragmented representations, limiting recommendation performance.

To address these issues, we propose DAMPS, a novel framework that performs fine-grained spectral domain multimodal representation calibration prior to multimodal representation fusion. First, to address the phase distribution shift across modalities, we introduce the adaptive phase calibrator for the first time in the multimodal recommendation scenario. In contrast to traditional phase alignment strategies in signal processing, which typically rely on static calibration, our phase calibrator adaptively rotates the spectrum of each modality to minimize phase spectrum shift. Second, to denoise the modality-specific amplitude spectrum, we design the amplitude variance-ratio filter that suppresses high-frequency fluctuations while preserving distinct spectral characteristics of each modality. Third, we propose the inter-modal coherence filter, which measures spectral coherence between modalities to selectively amplify the frequency components shared by both, thus reinforcing true joint preference information. Notably, DAMPS is model-agnostic and can be seamlessly integrated into existing multimodal recommendation pipelines. We validate DAMPS on four public Amazon datasets and one short-video dataset, and the results demonstrate its consistent ability to significantly enhance multimodal recommendation performance. Our contributions are summarized as follows:

- We propose DAMPS, a novel spectral domain multimodal representation calibration framework that operates prior to multimodal representation fusion for multimodal recommendation.
- We design the adaptive phase calibrator that synchronizes inter-modal phase distributions through the optimal minimum variance unbiased estimate of phase shift and adaptive phase rotation.
- We develop the amplitude variance-ratio filter to denoise the amplitude spectrum by suppressing high-frequency fluctuations while preserving modality-specific semantics.
- We introduce the inter-modal coherence filter that amplifies resonant frequency components shared between modalities to enhance the consistency of preference information.
- Extensive experiments on five real-world recommendation datasets demonstrate that DAMPS consistently improves multimodal recommendation performance and achieves state-of-the-art results.

2 Related Works

2.1 Multimodal recommendation

Multimodal recommendation frameworks seek to surpass the limitations of traditional recommendation methods by augmenting user-item interactions with visual, textual and other heterogeneous cues. Pioneering works such as VBPR [13], CKE [43] and DeepStyle [24] demonstrated the viability of this paradigm by leveraging pre-trained convolutional and language models to obtain modality-specific embeddings, which were subsequently concatenated with

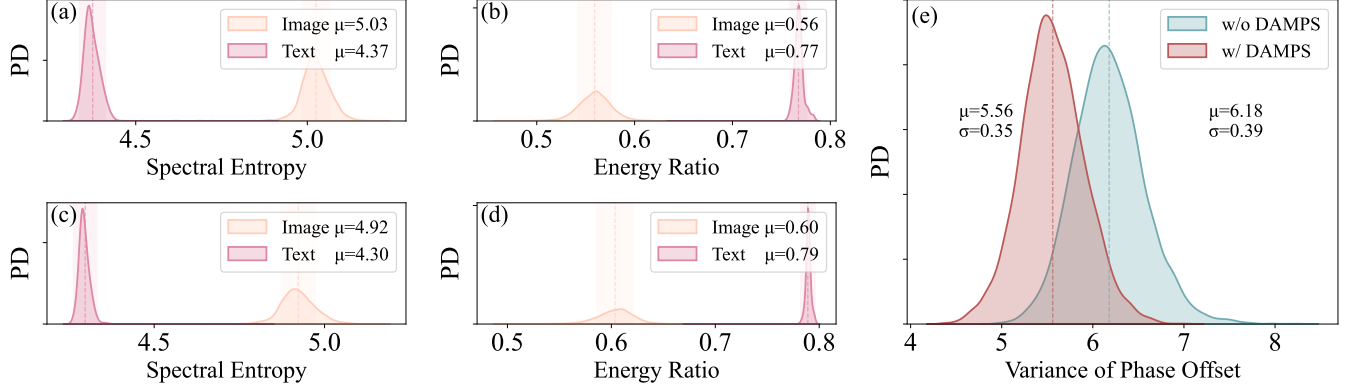


Figure 1: Uncertainty distributions of textual and visual representations in MGCN for Amazon Clothing dataset. (a)-(b) Original uncalibrated amplitude spectrum; (c)-(d) DAMPS-calibrated amplitude spectrum; (e) Variance of Phase Offset before/after DAMPS calibration. "PD" is the abbreviation for Probability Distribution.

latent user and item vectors to enrich preference information. Building on this foundation, attention-based architectures like ACF [7] and VECF [9] introduced cross-modal attention mechanisms that dynamically adjust the contribution of each modality, yielding more discriminative representations and improved recommendation performance. However, the effectiveness of simple embedding concatenation and fusion significantly declines in the presence of noise, highlighting the need for more robust fusion strategies.

Driven by the rapid advances of Graph Neural Networks [39], multimodal recommendation has progressively embraced toward spatial-domain propagation paradigms. LightGCN [14] laid the groundwork with a streamlined high-order message-passing framework between users and items, while MMGCN [37] mitigated multimodal noise by introducing separate modality-specific graphs and cross-modal gating. DualGNN [34] further amplified collaborative signals via alternating user-item and item-item propagation. LATITICE [44] dynamically learned item-level structures in each modality and injected high-order affinities into item representations through graph convolutions, and FREEDOM [48] further improved training stability by freezing the graph structures to curb redundancy. More recently, LGMRec [11] fused local and global attention priors before message passing to strengthen fine-grained multimodal alignment, whereas LIRDRec [49] integrated modality-specific encoders with a multimodal transformation module to explicitly capture cross-modal commonalities. Despite these notable achievements in the spatial domain, the exploration of spectral domain approaches remains embryonic. SMORE [26] pioneered multimodal fusion in the frequency domain by designing modality-specific dynamic filters tailored to spectral features. However, it only addresses denoising and has yet to resolve spectral inconsistencies that lead to distortion or loss of multimodal information.

Despite their remarkable performance, they have not simultaneously taken into account modality-specific noise and inter-modal distribution shifts. During multimodal information fusion, noise signals may inadvertently be amplified, thereby weakening the expressiveness of the fused representations and degrading recommendation performance. Distinct from all the aforementioned methods,

we propose to denoise and calibrate multimodal representations from the perspective of amplitude spectrum and phase spectrum in the spectral domain, thereby addressing these issues.

2.2 Uncertainty Estimation for Multimodal Data

Multimodal learning has advanced markedly by exploiting complementary signals, such as images, text, and even audio, to surpass the predictive power of any single modality [45]. Nevertheless, the reliability of input data remains a persistent bottleneck. While prevailing fusion methods emphasize inter-modal interaction and fusion, they typically neglect the intrinsic uncertainty introduced by noise and distribution shifts, thereby limiting their performance. To bridge this gap, recent studies have incorporated uncertainty estimation, quantifying distributional discrepancies between modalities and furnishing calibrated confidence scores that inform downstream decisions [10, 35]. Early efforts focused on modality-aware confidence weighting, followed by attention-calibration strategies that modulate inter-modal attention distributions based on predictive uncertainty, both yielding notable gains in tasks like visual question answering and emotion recognition [20, 31]. More recent evidence-theoretic fusion frameworks [32, 40] employ Dempster-Shafer theory [3] to reconcile conflicting cues while explicitly modeling belief quality. These insights are highly relevant to multimodal recommendation, where modal-specific noise and inter-modal distribution shifts similarly hinder performance. However, it is noteworthy that uncertainty modeling remains conspicuously underexplored in multimodal recommendation domain. Therefore, integrating systematic uncertainty estimation into multimodal recommendation frameworks holds promise for pioneering a new paradigm of confidence-aware and reliable recommendation. To this end, from the perspective of signal processing, we employ three metrics, Spectral Entropy, Energy Ratio, and Variance of Phase Offset, to estimate the uncertainty of multimodal representations. Subsequently, we reduce uncertainty in the spectral domain from both the amplitude spectrum and phase spectrum perspectives, in other words, denoising and calibrating multimodal representations.

3 Methodology

3.1 Problem Definition

In a multimodal recommendation system (MRS), let $\mathcal{U}$, $\mathcal{I}$, $\mathcal{E}$, and $\mathcal{M}$ represent the user set, item set, observed user-item interactions, and modality set, respectively. Numerically, the number of items is denoted by $N = |\mathcal{I}|$. In this work, we primarily consider two modalities: text and image, i.e., $\mathcal{M} = \{\text{text, image}\}$. Thus, in modality $m \in \mathcal{M}$, each item $i \in \mathcal{I}$ is represented by modality-specific feature vectors $\mathbf{x}_i^m \in \mathbb{R}^{d_m}$, where d_m denotes the feature dimension. Formally, we use $\mathcal{X}_m = \{\mathbf{x}_i^m \mid i \in \mathcal{I}\}$ to denote the set of features for all items under modality m , and define the full multimodal feature set as $\mathcal{X} = \{\mathcal{X}_m \mid m \in \mathcal{M}\}$. The objective of MRS is to train a robust recommendation model $r_\omega = \mathcal{W}(\mathcal{U}, \mathcal{I}, \mathcal{E}, g_{\mathcal{M}}(\mathcal{X}))$ that performs well on unobserved user-item interactions $\mathcal{E}'$ to achieve accurate personalized recommendations. Here, $g_{\mathcal{M}}: \mathcal{X} \mapsto \mathcal{Z}$ is the modality-aware feature projector, and $\mathcal{W}: \mathcal{Z} \mapsto \mathcal{E}'$ is the recommendation predictor, where $\mathcal{Z}$ represents the unified representation space.

From a signal processing perspective, multimodal signal fusion may inadvertently amplify modality-specific noise and introduce implicit semantic biases, thereby distorting inter-modal interactions. Therefore, our objective is to directly apply DAMPS to the *raw* multimodal item representations within the unified representation space, dynamically decoupling and calibrating their amplitude spectrum and phase spectrum components *without* altering the user-item interactions and the multimodal learning mechanisms in r_ω , in order to address these issues. Formally, $r_\omega^* = \mathcal{W}(\mathcal{U}, \mathcal{I}, \mathcal{E}, \mathcal{C}(g_{\mathcal{M}}(\mathcal{X})))$, where $\mathcal{C}: \mathcal{Z} \mapsto \mathcal{Z}$ denotes the calibration process of the *raw* multimodal representations in our DAMPS framework, and r_ω^* represents the multimodal recommendation model capable of delivering superior recommendations compared to the original model r_ω .

3.2 Framework Overview

Figure 2 illustrates the framework of DAMPS, which denoises and calibrates multimodal representations from both amplitude and phase perspectives in the spectral domain. Considering the distinct roles of the amplitude spectrum and phase spectrum, DAMPS consists of three modules to achieve multimodal representation calibration. Specifically, to address the issue of the phase distribution shift, we first propose the **Adaptive Phase Calibrator** to estimate the optimal phase offset between text and image representations at each frequency bin and apply adaptive rotation to enhance phase consistency. To suppress modality-specific noise, we introduce the **Amplitude Variance-Ratio Filter**, which evaluates the signal-to-noise ratio of the amplitude spectrum at each frequency bin to attenuate noise components while preserving prominent modality-specific features. To further mitigate residual distribution discrepancies, we design the **Inter-Modal Coherence Filter** to measure spectral coherence between modalities, thereby amplifying truly shared frequency components while suppressing noise artifacts.

3.3 Spectral Decomposition

Following the common strategy for processing multimodal features [26, 41, 49], we first project each uni-modal feature into a shared representation space $\mathbb{R}^d$ using Multi-Layer Perceptron

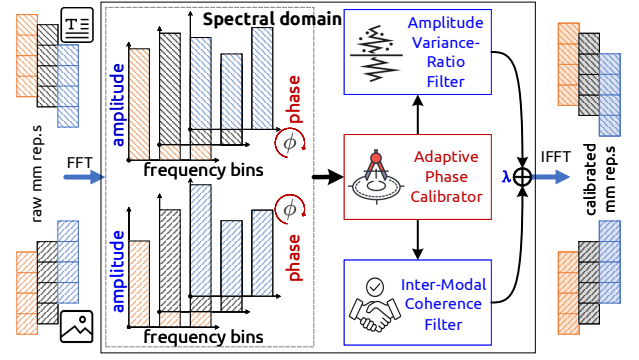


Figure 2: The overview of DAMPS framework. The multimodal representation calibrations in amplitude spectrum and phase spectrum are colored in blue and red, respectively.

(MLP). This ensures comparability among multimodal representations, enabling the estimation to reflect true modality shifts rather than arbitrary linear distortions. Specifically, given uni-modal features $\mathbf{x}_i^m$, we derive the latent uni-modal representation as follows:

$$\mathbf{h}_i^m = \mathbf{W}_m \mathbf{x}_i^m + \mathbf{b}_m \quad (1)$$

where $\mathbf{W}_m \in \mathbb{R}^{d \times d_m}$ and $\mathbf{b}_m \in \mathbb{R}^d$ are the learnable matrix and bias vector for each modality m , respectively.

Fast Fourier Transform. Although multimodal features are mapped into a unified representation space, Figure 1 reveals that the inter-modal distribution shifts and modality-specific intrinsic noise persist. To better denoise and calibrate multimodal representations, we first apply the F -point Fast Fourier Transform (FFT) to decompose uni-modal representation into its spectral-domain amplitude and phase components. Let $\mathbf{z}_i^m = \mathcal{F}_m(\mathbf{h}_i^m) \in \mathbb{C}^F$ denote the complex spectral-domain representation with $F = \frac{d}{2} + 1$ frequency bins. The amplitude and phase at each frequency bin f are given by:

$$\mathbf{z}_{i,f}^m = |\mathbf{z}_{i,f}^m| e^{j\phi_{i,f}^m}, \quad \text{where } \phi_{i,f}^m \in [-\pi, \pi), \quad (2)$$

with $|\mathbf{z}_{i,f}^m|$ and $\phi_{i,f}^m$ representing the amplitude and phase, respectively. Aggregating these components across all frequencies yields the amplitude spectrum and phase spectrum:

$$\mathbf{A}_i^m = [|\mathbf{z}_{i,1}^m|, \dots, |\mathbf{z}_{i,F}^m|]^\top, \quad (3)$$

$$\Phi_i^m = [\phi_{i,1}^m, \dots, \phi_{i,F}^m]^\top, \quad (4)$$

which collectively characterize the spectral properties of the item representation under modality m .

3.4 Adaptive Phase Calibrator

Based on the phase spectrum Φ_i^m obtained from spectral decomposition, we first perform inter-modal phase calibration. Specifically, to address the inherent lack of flexibility in traditional static phase alignment strategies, we propose the adaptive phase calibration strategy. This strategy dynamically adjusts phase rotation angle according to the differences in phase distributions across modalities, thereby precisely calibrating inter-modal phase distribution shifts. For each item $i \in \mathcal{I}$ and its associated text-image modality pair, we

compute its *phase offset vector* $\mathbf{R}_i \in \mathbb{R}^F$ at all frequencies to capture the systematic phase mismatch between modalities:

$$\mathbf{R}_i = \Phi_i^{\text{text}} - \Phi_i^{\text{image}} = \begin{bmatrix} \phi_{i,1}^{\text{text}} - \phi_{i,1}^{\text{image}} \\ \vdots \\ \phi_{i,F}^{\text{text}} - \phi_{i,F}^{\text{image}} \end{bmatrix}, \quad (5)$$

where each element is wrapped to $[-\pi, \pi)$.

Definition 3.1. (Vectorized MVU Circular Estimator). Given N items, each characterized by a phase offset vector $\mathbf{R}_i \in \mathbb{R}^F$ ($i = 1, \dots, N$), with a latent systematic bias $\boldsymbol{\theta} \in \mathbb{R}^F$, the minimum variance unbiased (MVU) estimator for $\boldsymbol{\theta}$ is the circular mean $\hat{\boldsymbol{\theta}}$:

$$\hat{\boldsymbol{\theta}} = \text{atan2} \left(\sum_{i=1}^N \sin \mathbf{R}_i, \sum_{i=1}^N \cos \mathbf{R}_i \right), \quad (6)$$

where $\sin$, $\cos$, and atan2 all perform element-wise operations.

The circular mean $\hat{\boldsymbol{\theta}}$ determines the dominant phase shift between modalities, and attains the Cramér-Rao lower bound [29] for circular distributions, with the detailed proof provided in Appendix B.1. To enhance the consistency between phases in the spectral domain, we use circular mean $\hat{\boldsymbol{\theta}}$ as the optimal rotation vector, and symmetrically adjust spectral representations for both modalities:

$$\begin{aligned} \mathbf{z}_i^{\text{image}} &\leftarrow \mathbf{z}_i^{\text{image}} e^{-j\frac{\hat{\boldsymbol{\theta}}}{2}}, \\ \mathbf{z}_i^{\text{text}} &\leftarrow \mathbf{z}_i^{\text{text}} e^{+j\frac{\hat{\boldsymbol{\theta}}}{2}}. \end{aligned} \quad (7)$$

Residual Phase Calibration with MVU Prior. Although calibration with the MVU can eliminate systematic phase shifts, it cannot fully address residual nonlinear deviations. By introducing trainable phase residual rotations, the framework can adaptively compensate for such residual deviations, thereby learning complex, data-dependent calibration patterns that static calibration cannot capture. Therefore, we introduce a learnable residual rotation vector $\boldsymbol{\psi} \in \mathbb{R}^F$, which can be optimized within the workflow of multimodal recommendation frameworks. Consequently, the complete phase calibration is formed by combining the MVU prior $\hat{\boldsymbol{\theta}}$ with the learnable residual $\boldsymbol{\psi}$ through element-wise operations.

$$\begin{aligned} \tilde{\mathbf{z}}_i^{\text{image}} &= \mathbf{z}_i^{\text{image}} e^{-j(\frac{\hat{\boldsymbol{\theta}}}{2} + \boldsymbol{\psi})}, \\ \tilde{\mathbf{z}}_i^{\text{text}} &= \mathbf{z}_i^{\text{text}} e^{+j(\frac{\hat{\boldsymbol{\theta}}}{2} + \boldsymbol{\psi})}. \end{aligned} \quad (8)$$

3.5 Amplitude Variance-Ratio Filter

Building upon the phase-calibrated spectral representation $\tilde{\mathbf{z}}_i^m$, our next goal is to disentangle true modality-specific spectral representations from spurious high-frequency noise in the amplitude spectrum. To achieve this, we first introduce the robust spectral variance decomposition to quantitatively distinguish modality-specific spectral representations from inherent noise components.

Robust Spectral Variance Decomposition. To ensure reliable noise estimation in the presence of potential outliers, we employ a robust statistical framework based on the median absolute deviation (MAD) [15]. We first collect the amplitudes of all items at each frequency bin f :

$$\tilde{\mathbf{A}}_{I,f}^m = [|\tilde{z}_{1,f}^m|, |\tilde{z}_{2,f}^m|, \dots, |\tilde{z}_{N,f}^m|]^\top, \quad (9)$$

where $I = \{1, \dots, N\}$ indexes the N items. Then, we compute the MAD at frequency bin f as:

$$\text{MAD}(\tilde{\mathbf{A}}_{I,f}^m) = \text{median}_{i \in I} \left| |\tilde{z}_{i,f}^m| - \text{median}_{j \in I} (|\tilde{z}_{j,f}^m|) \right|, \quad (10)$$

where the computation involves two key median operations over all the items I . First, the inner median calculates the central tendency of amplitude at frequency bin f across all items, establishing the robust estimate. Then, the outer median evaluates the typical deviation by computing the median of the absolute differences of all items's amplitudes relative to a central amplitude value.

To ensure our noise metric adheres to Gaussian statistical assumptions and provides an unbiased estimate of the standard deviation, we calibrate the MAD by multiplying it by the constant

$$c_{\text{mad}} = \frac{1}{\Phi^{-1}(3/4)} \approx 1.4826, \quad (11)$$

where $\Phi^{-1}(3/4) \approx 0.6745$ is the 75% of the standard normal distribution. So that under the Gaussian noise assumption

$$c_{\text{mad}} \text{MAD} \approx \sigma. \quad (12)$$

Having established robust noise estimation through MAD, the total variance can be computed by:

$$\sigma_{\text{total}}^2(\tilde{\mathbf{A}}_{I,f}^m) = \text{Var}_I [\tilde{\mathbf{A}}_{I,f}^m] = \mathbb{E}[|\tilde{\mathbf{A}}_{I,f}^m|^2] - |\mathbb{E}[\tilde{\mathbf{A}}_{I,f}^m]|^2 \quad (13)$$

where $\mathbb{E}$ denotes expectation over all items for each frequency bin f independently. Then, we decompose the total variance $\sigma_{\text{total}}^2(\tilde{\mathbf{A}}_{I,f}^m)$ into its constituent components: the intra-item variance $\sigma_{\text{intra}}^2(\tilde{\mathbf{A}}_{I,f}^m)$ and the inter-item variance $\sigma_{\text{inter}}^2(\tilde{\mathbf{A}}_{I,f}^m)$:

$$\sigma_{\text{intra}}^2(\tilde{\mathbf{A}}_{I,f}^m) = c_{\text{mad}}^2 [\text{MAD}(\tilde{\mathbf{A}}_{I,f}^m)]^2 \quad (14)$$

$$\sigma_{\text{inter}}^2(\tilde{\mathbf{A}}_{I,f}^m) = \max \left(\sigma_{\text{total}}^2(\tilde{\mathbf{A}}_{I,f}^m) - \sigma_{\text{intra}}^2(\tilde{\mathbf{A}}_{I,f}^m), 0 \right). \quad (15)$$

By utilizing the MAD-based intra-item variance $\sigma_{\text{intra}}^2(\tilde{\mathbf{A}}_{I,f}^m)$, the Amplitude Variance-Ratio Filter can reliably estimate the true intra-item noise, thereby preserving inter-item semantic energy to support accurate multimodal recommendation.

Variance-Ratio Filtering. After decomposing the total variance into signal and noise components, we define a smooth, bounded filter for modality m as

$$\mathbf{V}^m(\tilde{\mathbf{A}}_{I,f}^m) = \frac{\sigma_{\text{inter}}^2(\tilde{\mathbf{A}}_{I,f}^m)}{\sigma_{\text{inter}}^2(\tilde{\mathbf{A}}_{I,f}^m) + \sigma_{\text{intra}}^2(\tilde{\mathbf{A}}_{I,f}^m) + \varepsilon}, \quad \varepsilon > 0, \quad (16)$$

where ε denotes a small positive constant added to the denominator to ensure numerical stability by preventing division by zero (or extremely small values) when both $\sigma_{\text{intra}}^2(\tilde{\mathbf{A}}_{I,f}^m)$ and $\sigma_{\text{inter}}^2(\tilde{\mathbf{A}}_{I,f}^m)$ are near zero.

Definition 3.2 (Wiener Filter Equivalence). Let σ_{inter}^2 denote the inter-item variance (interpreted as signal power) and σ_{intra}^2 the intra-item variance (interpreted as noise power) [38]. The optimal minimum mean square error (MMSE) filter, i.e. the Wiener filter, is

$$\mathbf{H} = \frac{\sigma_{\text{inter}}^2}{\sigma_{\text{inter}}^2 + \sigma_{\text{intra}}^2}, \quad (17)$$

hence the amplitude variance-ratio filter coincides with the classical Wiener filter: it attenuates frequency components dominated

by noise while preserving those dominated by the true signal. The proof can be found in Appendix B.2.

Therefore, at frequency bin f , a larger value of $\mathbf{V}^m(\tilde{\mathbf{A}}_{T,f}^m)$ indicates this frequency is primarily dominated by stable semantic variation and should be preserved, while a smaller value suppresses noise-dominated components. By analyzing all frequency bins f through this criterion, the complete spectral-domain saliency representation $\mathbf{V}^m(\tilde{\mathbf{A}}_T^m) \in \mathbb{R}^F$ can be ultimately obtained. In practice, we treat $\mathbf{V}^m(\tilde{\mathbf{A}}_T^m)$ as the filter and apply it directly to the spectral representations:

$$\tilde{\mathbf{z}}_i^{m(\text{vrf})} = \mathbf{V}^m(\tilde{\mathbf{A}}_T^m) \odot \tilde{\mathbf{z}}_i^m, \quad m \in \{\text{text, image}\}. \quad (18)$$

where $\odot$ denotes the element-wise multiplication.

3.6 Inter-Modal Coherence Filter

Although the Amplitude Variance-Ratio Filter excels at distinguishing intra-modal noise from stable semantic representations, it overlooks a crucial fact that the high-order information about user interactions with items is often generated by the collaborative resonance of text and image at specific frequencies. In other words, a reliable reflection of user preference or contextual information between modalities can be achieved when the emphasis of textual keyword in the text and the prominence of visual features exhibit synchrony at specific frequencies. Therefore, to capture such interactions and resonance effects, we propose the Inter-Modal Coherence Filter that quantifies the inter-modal coupling strength via the cross-power spectrum.

Cross Power Spectrum. Let $\tilde{\mathbf{z}}_i^{\text{text}}$ and $\tilde{\mathbf{z}}_i^{\text{image}}$ denote the phase-calibrated representations for text and image of item i , respectively. We form the one-tap cross-power spectrum as

$$\mathbf{C}_i = \tilde{\mathbf{z}}_i^{\text{text}} \odot (\tilde{\mathbf{z}}_i^{\text{image}})^*, \quad (19)$$

where $(\cdot)^*$ denotes complex conjugation.

Amplitude-Squared Coherence. Next, we compute the power spectral density for each modality m to capture the energy in the amplitude spectrum, expressed as the square of the amplitude:

$$\mathbf{P}_i^m = |\tilde{\mathbf{z}}_i^m|^2, \quad m \in \{\text{text, image}\}. \quad (20)$$

Then, the Amplitude-Squared Coherence (ASC) can be defined as:

$$\text{ASC}_i = \frac{|\mathbf{C}_i|^2}{\mathbf{P}_i^{\text{text}} \mathbf{P}_i^{\text{image}}} \in [0, 1], \quad (21)$$

which assesses the linear dependence of two representations at amplitude spectrum [2, 12].

Functionally speaking, a higher ASC_i value indicates stronger inter-modal resonance and should be retained, while a lower value suggests uncorrelated noise, which can be used to suppress noise. Thus, we apply the coherence coefficients to filtering, in order to re-weight the spectral representations:

$$\tilde{\mathbf{z}}_i^{m(\text{coh})} = \text{ASC}_i \odot \tilde{\mathbf{z}}_i^m. \quad (22)$$

Finally, to enable DAMPS to dynamically balance noise suppression and inter-modal resonance for each modality representation,

we propose adaptively fusing them with a learnable vector $\boldsymbol{\lambda} \in \mathbb{R}^2$:

$$\begin{aligned} [\lambda_{\text{vrf}}, \lambda_{\text{coh}}] &= \text{softmax}(\boldsymbol{\lambda}), \\ \mathbf{z}_i^{m*} &= \lambda_{\text{vrf}} \tilde{\mathbf{z}}_i^{m(\text{vrf})} + \lambda_{\text{coh}} \tilde{\mathbf{z}}_i^{m(\text{coh})}. \end{aligned} \quad (23)$$

Thereafter, we employ the inverse Fast Fourier transform (IFFT) to project the denoised and phase-calibrated spectral representations back into the original representation space.

$$\mathbf{h}_i^{m*} = \mathcal{F}_m^{-1}(\mathbf{z}_i^{m*}) \quad (24)$$

Methodologically, given a multimodal recommendation model, we integrate the calibrated multimodal representations $\mathbf{h}_i^{m*}$ into the model architecture, instead of the original representations $\mathbf{h}_i^m$, to achieve superior multimodal recommendation performance.

4 Experiments

4.1 Experimental Settings

Datasets & Baselines. To thoroughly validate the effectiveness of our proposed DAMPS framework in improving multimodal recommendation performance, we conduct experiments on five publicly available datasets: four bimodal Amazon review datasets (*i*) Baby, *ii*) Sports and Outdoors, *iii*) Clothing, Shoes & Jewelry, and *iv*) Electronics, hereafter referred to as Baby, Sports, Clothing, and Electronics, and one trimodal dataset from a short-video platform, referred to as MicroLens. The detailed statistics of these datasets are shown in Table 1. Additionally, we choose seven state-of-the-art models published in the past three years as the backbone/baseline models and compare DAMPS against them: FREEDOM [48], BM3 [50], MGCN [41], LGMRec [11], SMORE [26], PGL [42], and LIR-DRec [49]. Since the majority of these baselines can only process data from two modalities, in this work we select only text and image modalities from the MicroLens dataset.

Table 1: Statistics of the datasets.

Dataset	#Users	#Items	#Interactions	Sparsity
Baby	19,445	7,050	160,792	99.88%
Sports	35,598	18,357	296,337	99.96%
Clothing	39,387	23,033	278,677	99.97%
Electronics	192,403	63,001	1,689,188	99.99%
MicroLens	98,129	17,228	705,174	99.96%

Evaluation Metrics & Implementation details. All experiments are conducted on a machine equipped with an NVIDIA 4090 GPU. The software environment includes Python 3.8.20 and PyTorch 2.4.1. We implement our proposed DAMPS framework and reproduce all backbone models based on the unified open-source MMRec framework [47]. It is specifically noted that DAMPS does not alter any structure of the backbone model, and is only embedded in a plug-in form. In addition, we adhere to the default hyperparameter settings suggested by each backbone and do not perform dedicated hyperparameter optimization on backbones integrated with DAMPS, in order to ensure statistical significance of DAMPS's performance. Following previous works [26, 47], we partition the user-item interactions into training, validation, and test sets in the ratio of

Table 2: Recall@K(R@K) and NDCG@K(N@K) performance comparison. Results are marked in bold when DAMPS yields performance improvements. Additionally, the best and runner-up results are highlighted in red and blue, respectively.

Dataset & Metrics	FREEDOM (MM'23)		BM3 (WWW'23)		MGCN (MM'23)		LGMRec (AAAI'24)		SMORE (WSDM'25)		PGL (AAAI'25)		LIRDRc (Arxiv'25)		Avg. Imp. (%)	
	raw	+Ours	raw	+Ours	raw	+Ours	raw	+Ours	raw	+Ours	raw	+Ours	raw	+Ours		
Baby	R@10	0.0624	0.0637	0.0556	0.0594	0.0615	0.0665	0.0644	0.0657	0.0673	0.0692	0.0621	0.0645	0.0663	0.0692	4.30
	R@20	0.0985	0.1008	0.0865	0.0915	0.0936	0.1003	0.0978	0.1032	0.1030	0.1088	0.0979	0.1006	0.1028	0.1061	4.63
	N@10	0.0324	0.0338	0.0293	0.0321	0.0329	0.0358	0.0353	0.0354	0.0366	0.0372	0.0332	0.0349	0.0367	0.0372	4.44
	N@20	0.0416	0.0430	0.0372	0.0404	0.0412	0.0445	0.0439	0.0450	0.0458	0.0473	0.0425	0.0441	0.0455	0.0467	4.59
Sports	R@10	0.0706	0.0754	0.0647	0.0737	0.0734	0.0797	0.0716	0.0738	0.0746	0.0789	0.0676	0.0711	0.0748	0.0765	6.51
	R@20	0.1083	0.1162	0.0992	0.1066	0.1108	0.1186	0.1058	0.1085	0.1142	0.1189	0.1040	0.1092	0.1134	0.1160	5.11
	N@10	0.0382	0.0415	0.0353	0.0413	0.0402	0.0440	0.0389	0.0402	0.0410	0.0432	0.0361	0.0387	0.0412	0.0420	7.56
	N@20	0.0479	0.0516	0.0442	0.0498	0.0498	0.0540	0.0477	0.0492	0.0512	0.0534	0.0455	0.0468	0.0510	0.0522	5.93
Clothing	R@10	0.0629	0.0690	0.0419	0.0498	0.0659	0.0735	0.0547	0.0589	0.0663	0.0714	0.0559	0.0611	0.0659	0.0679	9.68
	R@20	0.0931	0.1014	0.0629	0.0715	0.0961	0.1090	0.0805	0.0876	0.0988	0.1057	0.0847	0.0907	0.0979	0.0994	8.63
	N@10	0.0339	0.0373	0.0229	0.0281	0.0358	0.0402	0.0299	0.0325	0.0361	0.0387	0.0301	0.0331	0.0362	0.0372	10.52
	N@20	0.0416	0.0455	0.0282	0.0337	0.0435	0.0491	0.0365	0.0398	0.0443	0.0474	0.0374	0.0406	0.0445	0.0449	9.61
Electronics	R@10	0.0396	0.0432	0.0437	0.0482	0.0430	0.0470	0.0389	0.0411	0.0437	0.0464	0.0383	0.0412	0.0441	0.0457	7.39
	R@20	0.0601	0.0647	0.0648	0.0680	0.0636	0.0690	0.0587	0.0613	0.0645	0.0685	0.0575	0.0615	0.0667	0.0684	5.89
	N@10	0.0220	0.0238	0.0247	0.0276	0.0240	0.0263	0.0217	0.0229	0.0244	0.0259	0.0214	0.0230	0.0246	0.0254	7.42
	N@20	0.0273	0.0294	0.0302	0.0327	0.0293	0.0320	0.0268	0.0282	0.0298	0.0336	0.0263	0.0281	0.0304	0.0313	7.57
Microlens	R@10	0.0674	0.0746	0.0606	0.0743	0.0757	0.0855	0.0748	0.0782	0.0727	0.0767	0.0653	0.0740	0.0794	0.0827	10.54
	R@20	0.1032	0.1130	0.0981	0.1145	0.1134	0.1259	0.1132	0.1167	0.1121	0.1161	0.1025	0.1111	0.1185	0.1239	8.12
	N@10	0.0345	0.0386	0.0304	0.0388	0.0387	0.0444	0.0390	0.0412	0.0374	0.0395	0.0332	0.0387	0.0413	0.0430	12.31
	N@20	0.0437	0.0485	0.0400	0.0491	0.0484	0.0548	0.0489	0.0511	0.0475	0.0496	0.0428	0.0483	0.0514	0.0534	10.37

80%/10%/10%, respectively. To comprehensively evaluate the performance of DAMPS, we follow the full ranking protocol over all test users, and adopt Recall@K and NDCG@K as the evaluation metrics, with K taking values from {10, 20}.

4.2 Overall Performance

Table 2 summarizes the experimental results on the multimodal recommendation (MR) task. We observe that DAMPS consistently achieves performance improvements across all tasks and metrics for all backbone MR models, highlighting the superiority of our proposed DAMPS framework and the necessity of calibrating multimodal representations. Specifically, DAMPS achieves an average improvement across all metrics on five datasets, ranging from 4.49% on the Baby dataset to 10.34% on the Microlens dataset. The performance gains of DAMPS are particularly pronounced on image-dominated datasets (e.g., Clothing and Microlens), with improvements of at least 8.63% of Recall@20 on the Clothing dataset and up to 12.31% of NDCG@10 on the Microlens dataset across all metrics. These results strongly validate our core hypothesis: fine-grained spectral-domain multimodal representation calibration, achieved through coordinated amplitude spectrum denoising and phase spectrum consistency, effectively mitigates high-frequency noise in the visual modality while preserving the integrity of semantic content.

Most notably, DAMPS elevates the competitive ranking of MGCN from fourth place to first, which fully demonstrates the transformative potential of DAMPS. This advantage consistently extends to

other MR backbones, such as LGMRec, PGL, and LIRDRc. Remarkably, even when applied to SMORE, which already incorporates frequency-domain denoising, DAMPS still yields significant additional performance gains. This highlights the unique advantages of adaptive inter-modal phase consistency enhancement and more advanced amplitude denoising techniques in DAMPS over traditional spectral methods. In summary, DAMPS, as a plug-and-play enhancement framework, can be seamlessly integrated into existing MR models and significantly improve recommendation performance.

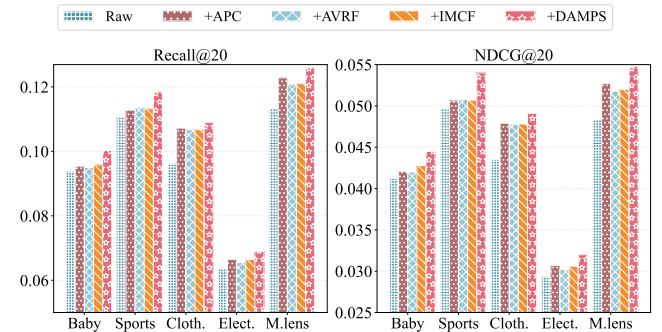


Figure 3: Performance comparisons in terms of Recall@20 and NDCG@20 on the benchmark datasets when each module of DAMPS is individually integrated with MGCN.

Table 3: Ablation study of the four most representative backbones across all datasets and metrics. Boldface indicates the best result for each metric under each backbone.

Dataset		BM3				MGCN				SMORE				LIRDRec			

4.3 Ablation Study

We conduct an ablation study to evaluate the impact of each key module in DAMPS. Specifically, we create three variants by removing a module from DAMPS, namely "without Adaptive Phase Calibrator"(-APC), "without Amplitude Variance-Ratio Filter"(-AVRF), and "without Inter-Modal Coherence Filter"(-IMCF). The performance comparison of DAMPS and its variants on four most representative backbones is shown in Table 3. We observe that the performance of DAMPS drops without APC, indicating that relying solely on amplitude calibration is far from sufficient. This validates the role of APC in mitigating potential semantic discrepancies between modalities. In addition, omitting AVRF leads to a performance decline in DAMPS. This confirms that preserving the unique characteristics of each modality while suppressing modality-specific noise is crucial for obtaining expressive multimodal representations and mitigating the negative effects of noise. Moreover, removing IMCF from DAMPS leads to a decrease in performance, which validates that leveraging inter-modal coherence can effectively capture inter-modal interaction patterns and is crucial for recommendation. In summary, removing any module from DAMPS weakens its expressiveness, demonstrating the importance of these modules.

To further evaluate the effectiveness of each of the three modules in DAMPS, we integrate each module individually into the backbones. Due to space constraints, only MGCN is utilized for illustration, as shown in Figure 3. The experimental results indicate that each module in DAMPS consistently enhances performance. Although each module already individually outperforms the backbone,

the complete DAMPS framework that integrates all three components achieves more superior performance. This demonstrates the necessity of simultaneously considering adaptive phase calibration, amplitude variance-ratio filtering, and inter-modal coherence filtering in DAMPS framework.

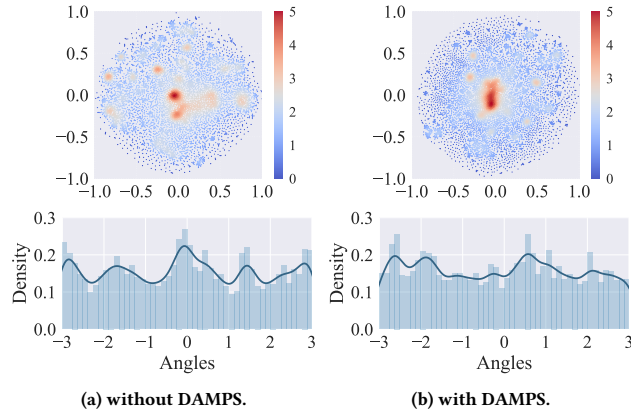
4.4 Visualization of Representations

To intuitively illustrate the effectiveness of the representations learned by backbones integrated with our DAMPS framework, we utilize t-SNE [33] to visualize the item representations generated by MGCN on the MicroLens dataset, both with and without DAMPS, as shown in Figure 4. We observe that, after multimodal representation calibration (MRC) in DAMPS, the phase distribution of representations with MRC in Figure 4(b) becomes markedly more uniform and stable than that of the raw representations without MRC in Figure 4(a), clearly underscoring the pivotal role of adaptive phase calibration in learning superior multimodal representations. Concurrently, the calibrated amplitude values accentuate the distinctive feature patterns of each modality, highlighting the necessity of robust high-frequency noise suppression prior to fusion. Moreover, the pronounced clusters of inter-modal resonance observed in the joint space confirm that exploiting inter-modal coherence is essential for capturing true user preferences. In summary, these findings validate that our multimodal representation calibration in DAMPS, which integrates adaptive phase calibration, amplitude variance-ratio filtering, and inter-modal coherence filtering

Table 4: Complexity and efficiency analysis. The values represent the percentage increase (Δ) in GPU memory, CPU memory, and training time after integrating DAMPS compared to the backbone.

Method	Baby			Sports			Clothing			Electronics			MicroLens		
	Δ GPU	Δ CPU	Δ Time	Δ GPU	Δ CPU	Δ Time	Δ GPU	Δ CPU	Δ Time	Δ GPU	Δ CPU	Δ Time	Δ GPU	Δ CPU	Δ Time
FREEDOM	+19.5%	+10.3%	+11.0%	+20.8%	+11.5%	+16.8%	+21.1%	+2.1%	+15.2%	+20.4%	+0.5%	+4.9%	+14.5%	+8.9%	+12.3%
BM3	+23.5%	+1.2%	+4.4%	+43.1%	+12.1%	+1.2%	+17.3%	+1.4%	+4.8%	+20.9%	+0.2%	+1.7%	+82.1%	+9.5%	+2.5%
MGCN	+0.1%	+0.1%	+2.6%	+0.1%	+0.8%	+3.7%	+0.5%	+0.4%	+3.6%	+1.0%	+0.2%	+6.3%	+5.6%	+3.8%	+1.2%
LGMRec	+28.6%	+5.4%	+37.8%	+37.3%	+8.3%	+47.0%	+39.8%	+1.0%	+77.1%	+28.6%	+0.3%	+5.1%	+12.9%	+6.2%	+20.7%
SMORE	+9.5%	+6.7%	+18.8%	+7.7%	+5.4%	+30.2%	+4.1%	+2.6%	+5.7%	+27.1%	+0.8%	+4.9%	+7.7%	+1.8%	+11.7%
PGL	+18.5%	+0.6%	+4.0%	+21.4%	+0.8%	+13.8%	+22.3%	+1.3%	+23.2%	+0.8%	+0.3%	+18.1%	+18.3%	+1.3%	+12.9%
LIRDRec	+22.0%	+0.8%	+5.6%	+27.8%	+0.6%	+45.0%	+29.8%	+0.9%	+32.9%	+56.8%	+0.2%	+31.3%	+8.9%	+0.5%	+25.8%

modules, yields more discriminative multimodal representations, ultimately enhancing multimodal recommendation performance.

**Figure 4: The t-SNE visualizations of items representations by MGCN on the MicroLens dataset without and with DAMPS.**

4.5 Complexity and Efficiency Analysis

We conduct a detailed analysis of the computational overhead to verify the efficiency of DAMPS. Theoretically, as an efficient plug-and-play module, DAMPS introduces only one additional FFT/IFFT operation and one element-wise rotation operation for the representation of each item. This results in a time complexity of $O(N'd(\log N'd))$ and a space complexity of $O(N'd)$, where N' represents the number of items in each batch, and d denotes the feature dimension, ensuring strong scalability.

To empirically substantiate this, we quantify the additional computational overhead introduced by DAMPS. Specifically, Table 4 reports the percentage increases (Δ) in GPU memory, CPU memory, and training time relative to the backbone models. The results demonstrate that DAMPS significantly enhances the performance of backbone models with acceptable additional overhead. Taking MGCN as an example, the resource overhead introduced by DAMPS is negligible. In terms of spatial overhead, GPU memory usage increases by only 0.1% to 5.6% across all datasets, and CPU memory usage rises by merely 0.1% to 3.8%. Regarding temporal overhead,

the increase in training time is modest, ranging between 1.2% and 6.3%. For the other backbones, although the average GPU memory usage increases by approximately 24%, the additional CPU memory overhead remains low, mostly staying below 5%. In terms of training efficiency, the increase in training time for all backbones falls within an acceptable range, with BM3 in particular showing less than a 5% increase. Overall, these empirical results indicate that DAMPS is a lightweight and computationally efficient solution.

5 Conclusion

In this paper, we have introduced DAMPS, a novel multimodal representation calibration framework for multimodal recommendation. We propose to simultaneously calibrate multimodal representations from the perspectives of amplitude spectrum and phase spectrum, thereby generating more expressive multimodal representations prior to fusion. Specifically, we propose the adaptive phase calibrator to calibrate phase spectrum discrepancies between different modalities. Furthermore, we design the amplitude variance-ratio filter to suppress modality noise while preserving modality-specific spectral features. Additionally, we introduce the inter-modal coherence filter to measure and enhance spectral coherence between modalities, thereby strengthening the truly shared information between modalities. Technically, DAMPS is model-agnostic and can be seamlessly integrated into existing multimodal recommendation workflows. Extensive experiments demonstrate that DAMPS consistently improves the recommendation performance, and achieves state-of-the-art results. Future work can extend our DAMPS framework to other multimodal domains and to more than two modalities.

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A Spectral Metrics Formulation

The modality representation vector $\mathbf{z}_i^m = \mathcal{F}_m(\mathbf{h}_i^m) \in \mathbb{C}^F$ is obtained by applying the Fast Fourier Transform (FFT) $\mathcal{F}_m$ to the original representation $\mathbf{h}_i^m$, where m denotes the modality (e.g., image or text) and F denotes the number of frequency bins. Based on this spectral representation, we perform three key quantitative evaluations. As illustrated by the probability distributions in Figure 1, each distribution is derived from the mean and standard deviation of the three spectral metrics.

A.1 Spectral Entropy Metric

To quantify the irregularity and complexity of spectral representations between modalities, we introduce the spectral entropy metric, which reflects the degree of energy dispersion in the spectral domain. Specifically, for each modality m , we compute a vector $H(\mathbf{z}^m) \in \mathbb{R}^N$, where each element corresponds to the entropy of the spectral energy distribution for a single item:

$$H(\mathbf{z}^m) = [H(\mathbf{z}_1^m), \dots, H(\mathbf{z}_i^m), \dots, H(\mathbf{z}_N^m)], \quad (25)$$

Each element $H(\mathbf{z}_i^m)$ is computed by:

$$H(\mathbf{z}_i^m) = - \sum_{k=1}^F P_{i,k} \log P_{i,k}, \quad (26)$$

$$P_{i,k} = \frac{|z_{i,k}^m|^2}{\sum_{j=1}^F |z_{i,j}^m|^2} \quad (27)$$

where $P_{i,k}$ represents the normalized power at the k -th frequency bin of $\mathbf{z}_i^m$. This measure evaluates the uniformity of energy distribution across frequency bins. Lower entropy values indicate more distinct modality-specific spectral patterns, while higher values reveal substantial noise contamination that disperses the feature distribution.

As shown in Figure 1, the observed mean spectral entropies ($\mu_{\text{image}} = 5.03$, $\mu_{\text{text}} = 4.37$) indicate substantial spectral randomness. We quantify noise dominance through normalized entropy ratios $\text{NoiseRatio}^m = (\mu_m / H_{\max}) \times 100\%$, where $H_{\max} = \log_2 N \approx 14.5$ bits (with $N = 23033$ represent the total number of items in the Clothing dataset). This yields concrete noise ratios of 34.7% (image) and 30.1% (text), meaning 32.4% of the modality feature dominated by noise.

A.2 Energy Ratio Metric

To assess the compactness of spectral energy distribution, we define the energy ratio metric, which measures the proportion of total energy concentrated in the most dominant frequency bins. For each modality m , we compute an energy ratio vector $R(\mathbf{z}^m) \in \mathbb{R}^N$, where each component represents the degree of energy concentration for an individual term:

$$R(\mathbf{z}^m) = [R(\mathbf{z}_1^m), \dots, R(\mathbf{z}_i^m), \dots, R(\mathbf{z}_N^m)] \quad (28)$$

$$R(\mathbf{z}_i^m) = \frac{\sum_{k \in \Omega} |z_{i,k}^m|^2}{\sum_{j=1}^F |z_{i,j}^m|^2} \quad (29)$$

where Ω contains the top 20% of frequency bins when sorted by amplitude [27]. This ratio ranges from 0.2 (perfectly uniform distribution) to 1.0 (all energy concentrated in the top bands), serving as an indicator of spectral sparsity. Higher values suggest the presence of strong, narrowband spectral features that typically correspond to meaningful signal components rather than broadband noise.

As shown in Figure 1, the low energy concentration ratios (0.56 for images, 0.77 for text) directly reveal substantial noise in the spectral representations. By defining the noise proportion as the energy deficit

$$1 - \text{mean}(R(\mathbf{z}^m)), \quad (30)$$

we obtain 44% noise for images and 23% for text, which means that roughly 33.5% of the spectral energy is devoted to uninformative or irrelevant features.

A.3 Phase Variation Metric

To evaluate inter-modal synchronization stability in the spectral domain, we propose a phase variation metric that quantifies the consistency of phase differences between modalities. For each item i , the phase offset at the k -th frequency bin between the text and image modalities is defined as:

$$\Delta\phi_{i,k} = \phi_{i,k}^{\text{text}} - \phi_{i,k}^{\text{image}}, \quad (31)$$

where $\phi_{i,k}^m$ denotes the phase of the complex spectral component at frequency k for modality m . We then compute the mean $\text{mean}(\Delta\phi_i)$ and variance $\text{var}(\Delta\phi_i)$ of the phase differences across all frequency bins:

$$\text{mean}(\Delta\phi_i) = \frac{1}{F} \sum_{k=1}^F \Delta\phi_{i,k} \quad (32)$$

$$\text{var}(\Delta\phi_i) = \frac{1}{F} \sum_{k=1}^F (\Delta\phi_{i,k} - \text{mean}(\Delta\phi_i))^2 \quad (33)$$

By collecting the variance values for all items, we define the phase variation vector $\text{var}(\Delta\phi) \in \mathbb{R}^N$ as:

$$\text{var}(\Delta\phi) = [\text{var}(\Delta\phi_1), \dots, \text{var}(\Delta\phi_N)] \quad (34)$$

This vector captures the degree of phase consistency between modalities for the entire dataset. A small $\text{var}(\Delta\phi_i)$ value indicates stable and consistent phase relationships (i.e., near-perfect phase locking) between the two modalities for item i , suggesting strong inter-modal synchronization. In contrast, larger values reflect more scattered or random phase differences, which may imply weak semantic alignment or modality mismatch in the spectral domain. The phase variation metric evaluates inter-modal synchronization stability through phase difference consistency.

B Proofs

B.1 Proof of Definition 1

PROOF. Let $\mathbf{R}_1, \dots, \mathbf{R}_N \in \mathbb{R}^F$ be independent circular observation vectors drawn from smooth, strictly positive densities $g_f(r_f - \theta_f)$ for each frequency bin f . We analyze the estimator's properties through its trigonometric moments:

Sufficient Statistics. For each frequency bin f , define the sufficient statistics:

$$\mathbf{C} = \sum_{i=1}^N \cos \mathbf{R}_i, \quad \mathbf{S} = \sum_{i=1}^N \sin \mathbf{R}_i, \quad (35)$$

where trigonometric functions operate element-wise. The circular mean estimator becomes:

$$\hat{\theta} = \text{atan2}(\mathbf{S}, \mathbf{C}), \quad (36)$$

which recovers the overall phase offset from these aggregated components.

Rotational Equivariance. A key property of phase estimation is that it must be invariant to the choice of coordinate system. We show that when all observed phases are rotated by a common δ , the estimator itself rotates by the same δ accordingly:

$$\begin{aligned} \mathbf{C} &\leftarrow \mathbf{C} \odot \cos \delta - \mathbf{S} \odot \sin \delta \\ \mathbf{S} &\leftarrow \mathbf{S} \odot \cos \delta + \mathbf{C} \odot \sin \delta, \end{aligned} \quad (37)$$

This transformation property ensures the estimator's covariance:

$$\hat{\theta} \mapsto \hat{\theta} + \delta, \quad (38)$$

This property is essential because it guarantees that our phase-difference estimates depend solely on the relative phase relationships between modalities, rather than on their absolute values. In multimodal analysis, this ensures that feature alignment remains objective and consistent.

Unbiasedness. To prove the estimator's unbiasedness, we examine the statistical properties of the trigonometric moments [51] of the phase offsets. For each frequency bin f , by leveraging the distribution's invariance under shifts, we obtain:

$$\mathbb{E}[\cos R_{i,f}] = A_f \cos \theta_f, \quad \mathbb{E}[\sin R_{i,f}] = A_f \sin \theta_f, \quad (39)$$

where $A_f = \int_0^{2\pi} \cos r g_f(r) dr > 0$ denotes the characteristic amplitude of the distribution. From this we can deduce:

$$\begin{aligned} \mathbb{E}[\hat{\theta}_f] &= \text{atan2}(\mathbb{E}[\mathbf{S}], \mathbb{E}[\mathbf{C}]) \\ &= \text{atan2}(NA \sin \theta_f, NA \cos \theta_f) \\ &= \theta_f. \end{aligned} \quad (40)$$

This result demonstrates that, although the atan2 function is nonlinear, aggregating a sufficiently large number of observations enables our estimator to precisely recover the true phase offset. Crucially, this property ensures that the calibration procedure in multimodal fusion does not introduce any systematic bias.

Optimality. The likelihood function has a separable structure:

$$L(\theta) = \prod_{f=1}^F \prod_{i=1}^N g_f(R_{i,f} - \theta_f), \quad (41)$$

which depends on the data only through $(\mathbf{C}, \mathbf{S})$, so by the Fisher–Neyman factorization theorem [18], $(\mathbf{C}, \mathbf{S})$ is sufficient for θ_f . In the special case of the von Mises family $g(\phi) \propto e^{\kappa \cos \phi}$, $(\mathbf{C}, \mathbf{S})$ is also complete. Hence, by the Lehmann–Scheffé theorem [5], any unbiased estimator that is a function of $(\mathbf{C}, \mathbf{S})$, in particular $\hat{\theta}_f$, is the unique minimum-variance unbiased estimator.

Finally, when each $R_{i,f}$ is von Mises, a straightforward calculation shows

$$\begin{aligned} I_1(\theta_f) &= \mathbb{E} \left[\left(\partial_{\theta_f} \log g(R_{i,f} - \theta_f) \right)^2 \right] \\ &= \kappa^2 \mathbb{E}[\sin^2(R_{i,f} - \theta_f)] = \kappa A, \end{aligned} \quad (42)$$

so the total Fisher information [25] is $I(\theta_f) = N\kappa A$, and the Cramér–Rao bound [28] gives

$$\text{Var}(\hat{\theta}_f) \geq \frac{1}{N\kappa A}. \quad (43)$$

A second-order Taylor expansion of the log-likelihood around θ_f , together with the delta method for $\text{atan2}(\mathbf{C}, \mathbf{S})$, shows equality holds. Therefore $\hat{\theta}_f$ attains the Cramér–Rao lower bound and is asymptotically efficient. $\square$

B.2 Proof of Definition 2

PROOF. Let $\mathbf{V}^m(\tilde{\mathbf{A}}_{I,f}^m) \in \mathbb{R}^F$ denote the vectorized variance-ratio filter for modality m . We prove its optimality through frequency-wise mean-square error analysis:

Error Decomposition. For each frequency bin f , the reconstruction error decomposes as:

$$\begin{aligned} \mathbb{E}[\|\mathbf{s} - \mathbf{V}^m(\tilde{\mathbf{A}}_{I,f}^m) \odot \tilde{\mathbf{z}}\|_2^2] &= \sum_{f=1}^F \mathbb{E} \left[(s_f - \mathbf{V}^m(\tilde{\mathbf{A}}_{I,f}^m)(s_f + \varepsilon_f))^2 \right] \\ &= \sum_{f=1}^F \left[(1 - \mathbf{V}^m(\tilde{\mathbf{A}}_{I,f}^m))^2 \sigma_{\text{inter}}^2(\tilde{\mathbf{A}}_{I,f}^m) \right. \\ &\quad \left. + (\mathbf{V}^m(\tilde{\mathbf{A}}_{I,f}^m))^2 \sigma_{\text{intra}}^2(\tilde{\mathbf{A}}_{I,f}^m) \right], \end{aligned} \quad (44)$$

where $\mathbf{s} = [s_1, s_2, \dots, s_F]$ represents the true semantic signal in the spectral domain, $\varepsilon = [\varepsilon_1, \varepsilon_2, \dots, \varepsilon_F]$ is the additive noise component, and we have used $\mathbb{E}[\mathbf{s}^2] = \sigma_{\text{inter}}^2(\tilde{\mathbf{A}}_{I,f}^m)$, $\mathbb{E}[\varepsilon^2] = \sigma_{\text{intra}}^2(\tilde{\mathbf{A}}_{I,f}^m)$, and $\mathbb{E}[\mathbf{s}\varepsilon] = 0$.

Optimality Condition. Taking the derivative with respect to the filter coefficient and setting it to zero:

$$\begin{aligned} \frac{\partial}{\partial \mathbf{V}^m(\tilde{\mathbf{A}}_{I,f}^m)} \left[(1 - \mathbf{V}^m(\tilde{\mathbf{A}}_{I,f}^m))^2 \sigma_{\text{inter}}^2(\tilde{\mathbf{A}}_{I,f}^m) \right. \\ \left. + (\mathbf{V}^m(\tilde{\mathbf{A}}_{I,f}^m))^2 \sigma_{\text{intra}}^2(\tilde{\mathbf{A}}_{I,f}^m) \right] = 0 \end{aligned} \quad (45)$$

yields the closed-form solution:

$$\mathbf{V}^m(\tilde{\mathbf{A}}_{I,f}^m) = \frac{\sigma_{\text{inter}}^2(\tilde{\mathbf{A}}_{I,f}^m)}{\sigma_{\text{inter}}^2(\tilde{\mathbf{A}}_{I,f}^m) + \sigma_{\text{intra}}^2(\tilde{\mathbf{A}}_{I,f}^m)} \quad (46)$$

Uniqueness. The Hessian matrix [8] is diagonal positive definite:

$$\nabla^2 = 2 \cdot \text{diag} \left(\sigma_{\text{inter}}^2(\tilde{\mathbf{A}}_{I,f}^m) + \sigma_{\text{intra}}^2(\tilde{\mathbf{A}}_{I,f}^m) \right) \succ 0 \quad (47)$$

is positive, this critical point is the unique global minimizer. Hence the amplitude variance-ratio filter indeed coincides with the Wiener filter [6] that achieves minimum MSE. $\square$